Assignment 3

A. Basic Conditionals (Simple)

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1. Positive / Negative / Zero

Write a program that asks the user to enter a number and prints whether the number is **positive**, **negative**, or **zero**.

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2. Age Group

Ask the user for their age. Based on the number entered, print whether they are a **child (below 13)**, **teenager (13–17)**, or **adult (18 and above)**.

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3. Temperature Check

Ask the user to enter the current temperature. If the temperature is **greater than 30°C**, print “Hot”, otherwise print “Cold”.

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4. Larger of Two Numbers

Ask the user to enter **two numbers**, and print which number is **larger**.

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5. Even or Odd

Ask the user to enter a number. Print whether the number is **even** or **odd**.

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6. Password Check

Ask the user to enter a password. If the entered password matches the predefined word `"python"`, print **“Access Granted”**, otherwise print **“Access Denied.”**

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7. Divisibility Check (Fizz/Buzz)

Ask the user for a number. If it is divisible by **5**, print “Fizz”. If it is divisible by **3**, print “Buzz”. (If both, print both messages.)

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8. Number in Range

Ask the user to enter a number and check if it lies within the range **1 to 10 (inclusive)**. Print whether it is **inside** or **outside** the range.

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9. Leap Year Check (Simple Rule)

Ask the user for a year and check if it is a **leap year**, using the simple rule: a year is leap if it is **divisible by 4**.

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10. Grade Based on Marks

Ask the user to enter their marks (out of 100). Print the grade according to:

- 90–100 → A
- 80–89 → B
- 70–79 → C
- Below 70 → Fail

B. Conditionals (Fun & Logical)

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11. Favourite Colour

Ask the user for their favourite colour. If they enter **"blue"**, print “Cool choice!”. Otherwise print “Nice!”

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12. Name Starting With A

Ask the user to enter their name. If the name begins with the letter **‘A’**, print “Awesome name!”, otherwise print “Nice name!”

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13. Is Number Too Big?

Ask the user for a number. If it is **greater than 100**, print “Too big!”, else print “Just right”.

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14. Equal Numbers Check

Ask the user for two numbers. Print **“Twins!”** if both numbers are equal; otherwise print **“Different Numbers!”**

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15. Height Eligibility

Ask the user for their height in centimeters. If it is **above 180 cm**, print “You can join the basketball team!”. Otherwise, print “You may try chess!”

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16. Divisible by 2 AND 3

Ask the user for a number and check if it is divisible by **both 2 and 3**. If yes → print “Special Number!”

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17. Check if Character is a Vowel

Ask the user to enter a **single character**. Check if it is a vowel (`a, e, i, o, u`) and print the result.

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18. Speed Limit Check

Ask the user for their driving speed. If the speed is **more than 80 km/h**, print “Overspeeding!” Otherwise, print “Safe Driving.”

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19. Check Round Number

Ask the user for a number. If its **last digit is 0**, print “Nice round number!”

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20. Check Multiple

Ask the user for two numbers. Print whether the **second number** is a **multiple** of the **first number**.

C. Basic Loops (Simple Repetition)

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21. Print Numbers 1–10

Write a program that prints all numbers from **1 to 10** using a loop.

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22. Even Numbers 1–20

Print all **even numbers** between **1 and 20** (inclusive).

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23. Reverse Counting 10–1

Print numbers from **10 down to 1** using a loop.

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24. Print 1 to n

Ask the user for a number **n**, and print numbers from **1 to n**.

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25. Multiplication Table of 5

Using a loop, print the **multiplication table of 5** (from 1×5 to 10×5).

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26. Multiplication Table for Any Number

Ask the user for a number and print its multiplication table from **1 to 10**.

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27. Print “Hello” 5 Times

Using a loop, print the word **"Hello"** exactly **5 times**.

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28. Print Characters of a Word

Ask the user for a word and print each character on a **new line** using a loop.

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29. Numbers Divisible by 3

Print all numbers between **1 and 30** that are divisible by **3**.

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30. Sum of Numbers 1–10

Calculate and print the **sum of numbers from 1 to 10** using a loop.

D. Loops + Conditionals (Combined Logic)

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31. Even Numbers up to n

Ask the user for a number **n**, and print all even numbers between **1 and n**.

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32. Squares of Numbers

Print the **square** of each number from **1 to 10**. (Example: 1→1, 2→4, 3→9 …)

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33. Count Multiples of 4

Count how many numbers between **1 and 20** are divisible by **4**, and print the count.

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34. Numbers Divisible by 2 and 5

Ask the user for a number **n**. Print all numbers between **1 and n** that are divisible by **both 2 and 5**.

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35. Numbers Ending with 7

Print all numbers between **1 and 50** whose **last digit is 7**. (Example: 7, 17, 27…)

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36. Multiples of 10

Print all multiples of **10** between **1 and 200**.

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37. Boom for Multiples of 3

Ask the user for a number **n**. Print all numbers from **1 to n**, but whenever a number is divisible by **3**, print **“Boom!”** instead of the number.

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38. Sum of Even Numbers up to n

Ask the user for a number **n**, and calculate the **sum of all even numbers** between **1 and n**.

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39. Print “Python!” n Times

Ask the user for a number **n** and print **“Python!”** exactly **n times**.

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40. Product of 1–5

Using a loop, find the **product** of numbers from **1 to 5**. (This is like factorial of 5.)

E. Fun and Challenging (Beginner-Friendly Logic)

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41. Guess the Secret Number

Create a guessing game: Keep asking the user to enter a number until they guess the **secret number** (a number you choose beforehand).

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42. Count Digits

Ask the user for a positive integer. Count how many **digits** the number has (e.g., 423 → 3 digits).

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43. Reverse a Number

Ask the user to enter a number. Reverse the number **using math only** (no strings). Example: 123 → 321

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44. Print a Star Pattern

Print the following pattern using loops:

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 *
**
***
```

(Three rows of increasing stars.)

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45. Odd Numbers Between Two Numbers

Ask the user for two numbers: **start** and **end**. Print all odd numbers between them (inclusive).

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46. Sum of Digits

Ask the user for a number and calculate the **sum of all its digits**. Example: 456 → 4+5+6 = 15

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47. Palindrome Check (Numbers)

Ask the user for a number. Check whether the number is a **palindrome** (same when reversed), using math only.

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48. Print n Stars in One Line

Ask the user for a number **n**, and print **n stars** on a single line. Example: n=5 → `*****`

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49. Countdown to Blast Off

Ask the user for a number **n**. Print a **countdown** from **n to 1**, and then print **"Blast off!"**

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50. Row Pattern (Row 1, Row 2…)

Ask the user for a number **n** and print the following pattern:

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 Row 1
Row 2
Row 3
...
Row n
```
